

Gauss-Markov Framework & Linear Models

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LSM

$$Y_{r \times 1} = X_{r \times p} \beta_{p \times 1} + \epsilon \quad \begin{aligned} E\epsilon &= 0 \\ \text{Var}\epsilon &= \sigma^2 I \end{aligned}$$

Estimation of β :

Least Squares estimate $G(\underline{b}) = \|Y - X\underline{b}\|_2^2$

$$\nabla G(\underline{b}) = \left(\frac{\partial G}{\partial b_j} \right)_{j=1}^p = 2X^T X \underline{b} - 2X^T Y$$

$$\nabla G(\underline{b}) = 0 \quad \leftarrow \text{Normal Equation}$$

ie, $X^T X \underline{b} = X^T Y$ $\leftarrow$ collection of p equations

Q1) Does it have a solution?

Q2) Is the solution unique?

Q3) If not then what are all the solⁿ?

A system of linear equations in variable x is given by

$$\underset{p \times q}{A} \underset{q \times 1}{x} = \underset{p \times 1}{y}$$

Solution exists if and only if $y \in \mathcal{C}(A)$

1. If $p=q$ and A is nonsingular then there is a unique solⁿ $\underline{x}^* = A^{-1}y$
2. If $Ax=y$ is solvable then Hy is a solution if H is a generalized inverse of A .

H satisfies

$$\underset{p \times q}{A} \underset{q \times p}{H} \underset{p \times q}{A} = \underset{p \times q}{A}$$

By rank factorization of A (or QR or SVD)

$A = PDQ$ where $P_{p \times p}$, $Q_{q \times q}$ are nonsingular

$$D = \underset{p \times q}{\begin{bmatrix} \Delta_{r \times r} & 0 \\ 0 & 0 \end{bmatrix}} \quad \Delta \text{ is diagonal with non-zero entries}$$

$$r = \text{rank}(A) \\ \leq \min\{p, q\}$$

$$\text{Take } H = \underset{q \times q}{Q}^{-1} \underset{q \times p}{D}^{-} \underset{p \times p}{P}^{-1} \text{ where } D^{-} = \begin{bmatrix} \Delta_{r \times r}^{-1} & 0 \\ 0 & 0 \end{bmatrix}$$

$$\begin{aligned} AHA &= PDQ \underset{q \times q}{Q}^{-1} D^{-} \underset{p \times p}{P}^{-1} PDQ \\ &= PDD^{-}DQ \\ &= PDQ \\ &= A \end{aligned}$$

Note: D^{-} is a g-inverse of D

Verify: Hy is a solution of $Ax = y$

$$\begin{aligned} AHy &= AHA \tilde{x} \quad (\text{since } y \in \mathcal{R}(A) \text{ for some } \tilde{x} \in \mathbb{R}^q) \\ &= A\tilde{x} \\ &= y \end{aligned}$$

Q) How to find all the solutions of $Ax=y$?

Note: If $\tilde{x}$ is a solⁿ then $\tilde{x} + \tilde{z}$ is also solⁿ, where $\tilde{z}$ is s.t. $A\tilde{z} = 0$

This means in particular that if $\text{rank}(A) = \# \text{ columns of } A$, then the null space of A is $\{0\}$ and hence solⁿ of $Ax=y$, if it exists, is unique

If A has full column rank and the system is solvable then the unique solⁿ is $A^{-1}y$ where $A^{-1}_{q \times p}$ is the left inverse of A i.e., it satisfies $A^{-1}A = I_q$

We denote a g -inverse of A by A^{-}

If H is a g -inverse of A , then

(i) HA is an idempotent matrix

check $HAHA = HA$

(ii) AH is an idempotent matrix

Remark: If $Ax=y$ is solvable and H is any g -inverse of A , then all the solⁿs of the system are of the form $Hyp + (HA - I)\tilde{z} \in \mathbb{R}^p$

First show that $(\checkmark)$ is a solution

$$A(Hy + (HA - I)\tilde{z}) = \underbrace{AH}_{=I}y + \underbrace{(AHA - A)}_0\tilde{z} = y$$

Next given a solⁿ say x^* , we want a z s.t.

$$x^* = Hy + (HA - I)z$$

Claim: Take $z = (HA - I)x^*$

$$\begin{aligned} Hy + (HA - I)z &= Hy + (HA - I)(HA - I)x^* \\ &= Hy + (I - HA)x^* \quad (\because HAH A = HA) \\ &= Hy + (I - HA)x^* \\ &= x^* + Hy - \underbrace{HAx^*}_y \\ &= x^* \end{aligned}$$

Remark: If A is symmetric matrix, then we can always find a symmetric g inverse

check Spectral Decomposition

$$A = P \Lambda P^T \quad \text{where } P^T P = P P^T = I_p$$

$p \times p \quad p \times p \quad p \times p \quad p \times p$

$$\Lambda = \begin{pmatrix} \Delta & 0 \\ 0 & 0 \end{pmatrix}$$

Δ is diagonal with non-zero diagonal elements

$$\text{Then } H = P \begin{pmatrix} \Delta^{-1} & 0 \\ 0 & R \end{pmatrix} P^T$$

$\underbrace{\hspace{1.5cm}}_{\text{arbitrary symm. matrix}}$

Least squares estimator of β

Any solⁿ of $X^T X \underline{b} = X^T Y$ — (★)
will be denoted by $\hat{\beta}$ and will be referred to as a least squares estimator of β .

(i) If $X^T X$ is non-singular $\Leftrightarrow X$ has full column rank, i.e. $\text{rank}(X) = p$, then $\hat{\beta}$ is unique and $\hat{\beta} = (X^T X)^{-1} X^T Y$

If X is not full column rank then

$\hat{\beta} = (X^T X)^{-} X^T Y$ is a least squares estimate.
where $(X^T X)^{-}$ is a (symmetric) g-inverse of $X^T X$.

Note: The class of all least squares estimate can be described by $(X^T X)^{-} X^T Y + ((X^T X)^{-} X^T X - I_p) z$
for $z \in \mathbb{R}^p$

Observe that $\hat{Y} = X \hat{\beta}$ is unique, even if $\hat{\beta}$ is not unique.

$$X \hat{\beta} = X (X^T X)^{-} X^T Y + [X (X^T X)^{-} X^T X - X] z \rightarrow 0$$

$$\rightarrow X^T = X^T X (X^T X)^{-} X^T \text{ (claim)}$$

$$\text{Note } \rho(X^T) = \rho(X^T X) \Rightarrow X^T = X^T X B$$

Since $(X^T X)^{-}$ is a g-inverse

$$X^T X (X^T X)^{-} X^T X = X^T X$$

multiply by B

Fact: $P_X \stackrel{\text{def.}}{=} X(X^T X)^- X^T$ is unique

Properties of P_X ① $P_X^T = P_X$

$$\begin{aligned} \text{② } P_X P_X &= X(X^T X)^- \underbrace{X^T X(X^T X)^-}_{X^T X} X^T \\ &= X(X^T X)^- X^T \\ &= P_X \end{aligned}$$

① and ② imply

P_X is an orthogonal projection matrix onto $\mathcal{L}(P_X)$

Take $v \in \mathbb{R}^n$

$$v = P_X v + (I - P_X)v \leftarrow \text{orthogonal}$$

Also if $v \in \mathcal{L}(P_X)$ then $P_X v = v$ decomposition

$\parallel ?$
 $\mathcal{L}(X)$ ($\mathcal{L}(P_X) \supset \mathcal{L}(X)$, obvious)

$$P_X = X(X^T X)^- X^T$$

Take $v = X \underline{b}$

$$\begin{aligned} \Rightarrow P_X v &= \underbrace{X(X^T X)^- X^T X}_{= X} \underline{b} \\ &= X \underline{b} \quad \therefore \mathcal{L}(P_X) \subset \mathcal{L}(X) \\ &= v \end{aligned}$$

Since P_X does not depend on the choice of $(X^T X)^-$
we may choose a very specific g -inverse, namely

the Moore-Penrose inverse of $X^T X$, denoted by $(X^T X)^+$ and then $P_X = X(X^T X)^+ X^T$

Construction

$$X^T X = U \Lambda U^T = U \begin{pmatrix} \Delta_{r \times r} & 0 \\ 0 & 0 \end{pmatrix} U^T$$

$$U^T U = U U^T = I_p$$

Δ is diagonal with positive diagonal elements

$$\text{Then } (X^T X)^+ = U \begin{pmatrix} \Delta^{-1} & 0 \\ 0 & 0 \end{pmatrix} U^T$$

What is P_X ?

$$X(X^T X)^+ X^T = X U \begin{pmatrix} \Delta^{-1} & 0 \\ 0 & 0 \end{pmatrix} U^T X^T$$

In particular if $X = V D U^T$ when $V^T V = V V^T = I_n$

Exe. $X U = V D$

columns of $X U$ are mutually orthogonal

$$D = \begin{pmatrix} \Delta^{1/2} & 0 \\ 0 & 0 \end{pmatrix}$$

$$\Delta^{1/2} = \text{diag} \begin{pmatrix} \sqrt{\delta_1} & 0 \\ 0 & \ddots \sqrt{\delta_p} \end{pmatrix}$$

$$= V D \begin{pmatrix} \Delta^{-1} & 0 \\ 0 & 0 \end{pmatrix} D^T V^T$$

$$= V \begin{pmatrix} I_r & 0 \\ 0 & 0 \end{pmatrix} V^T$$

Is there a solution of $X^T X \underline{b} = X^T Y$? YES.

- ① Solution of * always exists
- ② $(X^T X)^- X^T Y + ((X^T X)^- X^T X - I) \underline{z}$ ← all solutions
- ③ $\hat{Y} = X^T \hat{\beta}$ is unique
 $= P_X Y$ and P_X is the orthogonal projector $\mathcal{C}(X)$

Defⁿ: A linear parametric function $\underline{a}^T \underline{\beta}$ is estimable
 if $\exists \underline{l} \in \mathbb{R}^n$ s.t. $E(\underline{l}^T Y) = \underline{a}^T \underline{\beta}$ for all $\underline{\beta}$.

$$\Leftrightarrow \exists \underline{l} \text{ s.t. } \underline{a} = X^T \underline{l} \Leftrightarrow \underline{a} \in \mathcal{C}(X^T) = \mathcal{R}(X).$$

Ex. Suppose $Y_{ij} = \alpha_i + \gamma_j + \epsilon_{ij} \quad \begin{matrix} i=1,2 \\ j=1,2 \end{matrix}$

$$\underline{\beta} = \begin{pmatrix} \alpha_1 \\ \alpha_2 \\ \gamma_1 \\ \gamma_2 \end{pmatrix}$$

ϵ_{ij} have 0 mean
 (and equal variance)
 (and independent)

$$\alpha_1, \alpha_2, \gamma_1, \gamma_2 \in \mathbb{R}$$

$$\underline{a}^T \underline{\beta} = a_1 \alpha_1 + a_2 \alpha_2 + a_3 \gamma_1 + a_4 \gamma_2$$

$$\begin{pmatrix} Y_{11} \\ Y_{12} \\ Y_{21} \\ Y_{22} \end{pmatrix} = \begin{pmatrix} 1 & 0 & 1 & 0 \\ 1 & 0 & 0 & 1 \\ 0 & 1 & 1 & 0 \\ 0 & 1 & 0 & 1 \end{pmatrix} \begin{pmatrix} \alpha_1 \\ \alpha_2 \\ \gamma_1 \\ \gamma_2 \end{pmatrix} + \begin{pmatrix} \epsilon_{11} \\ \epsilon_{12} \\ \epsilon_{21} \\ \epsilon_{22} \end{pmatrix}$$

$$\alpha_{i1} + \alpha_{i2} = \gamma_{i3} + \gamma_{i4}$$